Protest Committee Report

Regattatron race `4ec6514d-62de-4858-a1b1-09b7c46e99e8` · Charleston (Crab Bank) · Windward–leeward, 1 lap, 11 boats
Incident: leeward gate, port-hand mark (`LWP`) · Boats: **AidanGoldfarb** (slot 0, finished 4th) and **Sully** (slot 1, finished 8th)
Question referred: *Did Sully have an overlap when the boats reached the zone at the leeward mark?*
Prepared 20 October 2026 by the ProtestCommittee toolkit (deterministic replay analysis) at the request of Aidan Goldfarb. Rules: Racing Rules of Sailing 2025–2028.

> **Units.** The replay reports positions in game units that its schema labels metres. The boat sprite is 24 of those units long, so the game's unit is not a real metre for the class being modelled. Overlap geometry is scale-free, so every distance in this report is given in game units (u) and in hull lengths (BL, 1 BL = 24 u). Where a real-world figure is useful, the Laser hull length of 4.23 m is used: 1 u ≈ 0.18 m, 1 BL = 4.23 m. The decision does not depend on the conversion.

Decision

No. At the instant AidanGoldfarb's bow reached the zone, Sully was **clear astern**: her bow was behind the line abeam AidanGoldfarb's aftermost point by 1.9 game units, which is 0.08 of a hull length, about 0.33 m (one foot) on a Laser. The boats were therefore not overlapped when the first of them reached the zone, and under rule 18.2(a)(2) Sully, the boat that had not reached the zone, was required to give AidanGoldfarb mark-room. The overlap that Sully later held began about two seconds after zone entry, when AidanGoldfarb bore away toward the mark; under rule 18.2(a) a later overlap does not change the obligation. Any residual doubt is resolved against Sully by rule 18.2(e).

Throughout the approach and rounding AidanGoldfarb was the leeward boat on the same tack and held right of way under rule 11. If AidanGoldfarb broke rule 16.1 while rounding, she is exonerated by rule 43.1(b) because she was sailing within the mark-room to which she was entitled.

The engine's original charge against AidanGoldfarb (rule 16, t = 2736 s) is not sustained on these facts. The engine's later reversal charging Sully is correct in outcome but cites the wrong rule: Sully was never the right-of-way boat, so rule 16 could not apply to her; the applicable rules were 18.2(a)(2) and 11.

1. Facts found

Times are the replay's simulation clock in seconds (sample index equals seconds; the start signal was at 1840.1 s). Positions are metres in the course frame, +y toward the windward mark. Headings are compass degrees. Wind was from 9° to 17° during the approach, veering steadily. Both boats were on port tack from t = 2709 to the rounding.

Fact	Finding
Mark and zone	Port gate mark `LWP` at (105.8, −800). Zone radius 100.8 u (4.2 BL) as defined by the game (`zoneRadiusM`).
Hull dimensions (from the game's renderer)	Hull 24 u long, 10 u beam (25 × 11 u including outline). The bow is 14 u (0.58 BL) ahead of the boat's position point; the stern is 10 u (0.42 BL) behind it. On a Laser that is a 4.23 m hull with the position point 2.47 m from the bow. The zone is therefore about 4.2 hull lengths, not the RRS 3; the game uses its own radius.
Zone entry	AidanGoldfarb's bow reached 100.8 u (4.2 BL) from the mark at **t ≈ 2718.8 s** (linear interpolation of samples 2718–2719). The replay viewer, bisected to the same bow distance, shows the frame at its clock 2718.5 s; the viewer's clock sits about 0.25 s off the sample clock. Sully's bow reached the zone at t ≈ 2720.
Overlap at zone entry	**Not overlapped.** Sully's bow was 1.9 u behind the line abeam AidanGoldfarb's aftermost point: 0.08 BL, about 0.33 m or one foot on a Laser (headings 146.7° and 141.9°). Sully had been clear astern by 0.08–0.16 BL continuously from t = 2709.
When the overlap began	Between t = 2720 and 2721, when AidanGoldfarb bore away from 146.7° to 136.7° toward the mark. The bear-away rotated her stern line and put Sully's bow 3.9 u (0.16 BL, about 0.7 m on a Laser) ahead of it. Sully held 141.9°.
Relative position	Sully was 36–38 u (about 1.5 BL) to windward (north) of AidanGoldfarb and on the inside of the rounding. AidanGoldfarb was the leeward boat.
Rounding	From t = 2731 AidanGoldfarb turned left from 117° to 52° at about 10°/s; Sully turned from 132° to 62° starting one second later. Closest approach t = 2735, position points 11.5 u apart, roughly one beam (0.48 BL). Contact cannot be confirmed or excluded from positions alone.
Engine events	t = 2736.0: INFRACTION, AidanGoldfarb, rule 16, "Not enough time and space", counterparty Sully. t = 2801.5: EXONERATION AidanGoldfarb and, at the same instant, INFRACTION Sully, rule 16. t = 2805–2820: Sully completes a One-Turn Penalty (360° heading sweep at under 3 kt). t = 2820.5: EXONERATION Sully.
Afterwards	AidanGoldfarb tacked to starboard at t = 2769–2775 (a tack, not a penalty turn) and did not pass head to wind or leave the zone before the rounding was complete.

1.1 Sample window around zone entry

All distances in game units (24 u = 1 hull length; on a Laser 1 u ≈ 0.18 m). Margin is the distance of Sully's bow ahead (+) or behind (–) the line abeam AidanGoldfarb's stern, measured along AidanGoldfarb's heading, using the game's hull offsets (bow +14 u, stern −10 u). Bow→mark is AidanGoldfarb's bow-tip distance to `LWP`.

t (s)	wind from	AidanGoldfarb x, y	hdg	kt	Sully x, y	hdg	kt	bow→mark (u)	margin (u)	state
2716	11.2°	−9.68, −738.81	146.7	13.9	−2.52, −702.30	141.9	14.5	118.6	−2.63	clear astern
2717	11.5°	−5.52, −745.91	146.7	14.0	2.29, −709.27	141.9	14.5	112.0	−2.39	clear astern
2718	11.8°	−1.36, −753.04	146.7	14.0	7.10, −716.25	141.9	14.5	105.5	−2.15	clear astern
2718.8	12.1°	**AidanGoldfarb's bow reaches the zone (interpolated)**						**100.8**	**−1.9**	**clear astern**
2719	12.2°	2.81, −760.17	146.7	14.0	11.92, −723.24	141.9	14.5	99.4	−1.91	clear astern
2720	12.5°	7.00, −767.32	146.7	14.1	16.75, −730.23	141.9	14.5	93.5	−1.70	clear astern; Sully's bow enters zone
2721	12.8°	11.94, −773.91	136.7	14.1	21.58, −737.23	141.9	14.6	85.7	+3.86	overlapped (AidanGoldfarb bore away)
2722	13.1°	17.11, −780.28	136.7	14.1	26.42, −744.24	141.9	14.6	79.7	+4.10	overlapped
2724	13.8°	27.68, −792.80	131.7	14.1	36.13, −758.30	141.9	14.6	67.7	+7.14	overlapped
2726	14.4°	40.14, −802.82	116.7	14.0	45.94, −772.49	141.9	14.9	53.9	+14.22	overlapped

1.2 Sensitivity of the overlap finding to the hull model

The only modelling input that can change the answer is where the bow and stern sit relative to the boat's recorded position point. Three models are compared at the instant of zone entry.

Hull model	Bow offset	Stern offset	Margin at zone entry	In hull lengths	Finding
Game renderer sprite (measured; used for this decision)	+14.0 u	−10.0 u	−1.9 u	−0.08 BL	clear astern
Symmetric 25 u hull	+12.5 u	−12.5 u	−0.9 u	−0.04 BL	clear astern
Symmetric 33.6 u hull (zone ÷ 3; the toolkit's prior default)	+16.8 u	−16.8 u	+7.7 u	+0.23 BL	overlapped

The game's zone is not three hull lengths, so the prior default was wrong for this game. The measured sprite governs. The decision would reverse only if the rules engine modelled a hull about 8% longer than the one it draws, which the renderer contradicts.

2. Diagrams: the game's replay at the instant of zone entry

Both images are captures of Regattatron's own replay viewer, paused at the frame in which AidanGoldfarb's bow tip lies on the zone circle (found by bisection on the viewer's interpolated positions). The red line is drawn through AidanGoldfarb's aftermost point, perpendicular to her heading, from the viewer's own boat coordinates; nothing is re-drawn. White dot: AidanGoldfarb's bow tip on the zone circle. Red dot: her aftermost point. Green dot: Sully's bow.



Figure 1. Zone-entry instant, zoomed. Sully's bow is 1.9 u (0.08 BL, about a foot on a Laser) on the astern side of the line abeam AidanGoldfarb's stern. The on-screen label uses the replay's raw units.



Figure 2. Same instant, full frame: both gate marks with their 100.8 u (4.2 BL) zones, camera following AidanGoldfarb. Wind from the top of the frame.

3. Rules applied

Quoted from the Racing Rules of Sailing 2025–2028 as published at racingrulesofsailing.org (see References).

Definitions

> **Clear Astern and Clear Ahead; Overlap.** One boat is clear astern of another when her hull and equipment in normal position are behind a line abeam from the aftermost point of the other boat's hull and equipment in normal position. The other boat is clear ahead. They overlap when neither is clear astern. However, they also overlap when a boat between them overlaps both. These terms always apply to boats on the same tack. They apply to boats on opposite tacks only when rule 18 applies between them or when both boats are sailing more than ninety degrees from the true wind.

> **Zone.** The area around a mark within a distance of three hull lengths of the boat nearer to it. A boat is in the zone when any part of her hull is in the zone.

> **Mark-Room.** Room for a boat (a) to sail to the mark when her proper course is to sail close to it, (b) to round or pass the mark on the required side, and (c) to leave it astern.

> **Keep Clear.** A boat keeps clear of a right-of-way boat (a) if the right-of-way boat can sail her course with no need to take avoiding action and, (b) when the boats are overlapped, if the right-of-way boat can also change course in both directions without immediately making contact.

Rule 18.2 Giving Mark-Room

> (a) When the first of two boats reaches the zone, (1) if the boats are overlapped, the outside boat at that moment shall give the inside boat mark-room; (2) if the boats are not overlapped, the boat that has not reached the zone at that moment shall give the other boat mark-room. When a boat is required to give mark-room by this rule, she shall continue to do so for as long as this rule applies, even if later an overlap is broken or a new overlap begins.
> (b) Rule 18.2(a) no longer applies if the boat entitled to mark-room passes head to wind or leaves the zone.
> (c) When rule 18.2(a) does not apply and the boats are overlapped, the outside boat shall give the inside boat mark-room.
> (d) If a boat obtained an inside overlap from clear astern or by tacking to windward of the other boat and, from

> the time the overlap began, the outside boat has been unable to give mark-room, rules 18.2(a) and 18.2(c) do not apply between them.
> (e) If there is a reasonable doubt that a boat obtained or broke an overlap in time, it shall be presumed that she did not.

Other rules relied on

- **Rule 11** (On the Same Tack, Overlapped): when boats are on the same tack and overlapped, the windward boat shall keep clear of the leeward boat. Sully was windward throughout.
- **Rule 16.1** (Changing Course): when a right-of-way boat changes course she shall give the other boat room to keep clear. It binds the right-of-way boat only.
- **Rule 43.1(b)** (Exoneration): a boat that breaks a rule of Section A, rule 15, 16 or 31 while sailing within the room or mark-room to which she is entitled is exonerated.
- **Rule 18.1**: rule 18 applied because both boats were required to leave the mark on the same side and were on the same tack while approaching it.

Application

1. Rule 18 applied from the moment the first boat reached the zone (rule 18.1). That boat was AidanGoldfarb, at t ≈ 2718.8.
2. At that moment Sully's hull was behind the line abeam AidanGoldfarb's aftermost point by 0.08 of a hull length (1.9 u; about a foot on a Laser). By the definition, Sully was clear astern and the boats were not overlapped.
3. Rule 18.2(a)(2) therefore applied: Sully, the boat that had not reached the zone, was required to give AidanGoldfarb mark-room, and to continue doing so even though an overlap began about two seconds later.
4. A foot is a fine margin. If it were regarded as doubtful, rule 18.2(e) presumes that Sully did not obtain the overlap in time.
5. AidanGoldfarb did not pass head to wind or leave the zone before completing the rounding, so rule 18.2(a) continued to apply (rule 18.2(b)).
6. AidanGoldfarb was the leeward, right-of-way boat under rule 11 and was entitled to mark-room. Her bear-away to the mark and her turn around it were sailing to the mark and rounding it on the required side, which is the content of mark-room. If that course change deprived Sully of room to keep clear, AidanGoldfarb is exonerated by rule 43.1(b).
7. Sully, as the windward boat not entitled to mark-room, was required to keep clear (rule 11) and to give mark-room (rule 18.2(a)(2)). Whether she in fact failed to keep clear, and whether there was contact, cannot be determined from position data alone (see Limits).

4. World Sailing Case Book 2025–2028: cases considered

Case numbers refer to the World Sailing Case Book 2025–2028 (edition v2025-07). Summaries are paraphrased; the authoritative text is in the Case Book itself.

Case	Rules	Relevance
Case 2	12, 14, 15, 18.2(a)(2), 43.1(c)	Directly on point. A boat that reaches the zone first while the other is not overlapped is owed mark-room by the boat that has not yet reached the zone, and a later change in overlap does not alter that. The boat entitled to mark-room is exonerated for Section A breaches while sailing within it.
Case 12	11, 14, 18.1, 18.2(a)(1), 43	Establishes that the overlap test is applied at the instant the first boat reaches the zone, regardless of how different the boats' courses are. Applied here, it fixes the test at t ≈ 2718.8, when the boats were not overlapped.
Case 118	18.2(a)(1), definitions Mark-Room, Room, Proper Course	Defines the extent of mark-room: space to sail promptly and in a seamanlike way to a position close to the mark on the required side, round it, and leave it astern. Used to characterise AidanGoldfarb's bear-away and rounding as sailing within her mark-room.
Case 114	14, 16.1, 18.2(a)(1), 31	Room to which a boat is entitled includes the space she needs to comply with her other obligations under Part 2 and rule 31.
Case 59	18.2(a)(2)	A boat outside the zone that changes course toward the mark and thereby becomes overlapped with a boat already in the zone does not acquire mark-room rights; the boat in the zone keeps hers.
Case 63	18.2(a), 43.1(b)	When space is made available at a mark to a boat not entitled to it, she may take it only at her own risk. Relevant to Sully's inside position after the overlap began.
Case 25	11, 14, 16.1, 18, 43	After mark-room has been given rule 18 ceases but rule 11 continues; an inside windward boat must still keep clear of the outside leeward boat. Confirms rule 11 governed the two boats after the rounding.

5. Audit of the engine's calls

t (s)	Engine event	Committee assessment
2736.0	INFRACTION, AidanGoldfarb, rule 16, "Not enough time and space", vs Sully	**Not sustained.** AidanGoldfarb was the right-of-way boat and did change course, so rule 16.1 was engaged, but she was sailing within the mark-room she was owed under rule 18.2(a)(2) and is exonerated by rule 43.1(b). The engine appears to have treated Sully as the inside boat entitled to mark-room, which requires an overlap at zone entry that the positions do not show.
2801.5	EXONERATION, AidanGoldfarb (no penalty turn was sailed)	Consistent with the committee's finding. The mechanism is unclear: the event coincides with the new charge against Sully, which suggests a re-adjudication or a player protest flow rather than a penalty taken. To be confirmed with the author.
2801.5	INFRACTION, Sully, rule 16	**Right boat, wrong rule.** Rule 16 binds only a right-of-way boat; Sully was the windward keep-clear boat throughout. The rules she could have broken are 18.2(a)(2) and 11.
2805–2820.5	Sully's One-Turn Penalty and EXONERATION	A penalty was due from Sully if she failed to give mark-room or keep clear. The committee cannot confirm a breach from positions alone (no contact model), but the obligation was hers.

6. Method

- **Data.** The replay blob (schema 2, 3472 samples at 1 Hz, 27 engine events) was read from the debrief page's React component state and the sample window 2640–2900 extracted for slots 0 and 1 and any boat within 200 u. Conventions as verified in the toolkit methodology: metres, +y upwind, compass headings, wind direction given as the direction it blows from, position point as recorded by the game.
- **Hull model.** Read from the replay viewer's PixiJS scene: the boat container's hull graphic has local bounds $x \in [-5, 5]$, $y \in [-14, +10]$ (outline ±5.5, −14.5…10.5), with the negative-y end the bow. The container's position is the sample position and its rotation is the heading. Hence bow offset +14 u, stern offset −10 u, beam 10 u, hull length 24 u. All overlap arithmetic is done in these units and reported as fractions of the hull length, so the real size of the class does not enter the decision.
- **Zone entry.** The instant at which the bow tip's distance to `LWP` equals 100.8 u, found two ways: linear interpolation between samples 2718 and 2719 (t = 2718.77) and bisection on the viewer's interpolated positions (viewer clock 2718.52). Both give the same geometry.
- **Overlap test.** Sully's bow point was projected onto AidanGoldfarb's heading axis and compared with the position of AidanGoldfarb's aftermost point (the "line abeam"). Negative margin means clear astern. Equipment in normal position is not modelled; the game draws none beyond the hull.
- **Right of way.** Tack from heading relative to wind; windward boat from the projection of the separation vector onto the wind-from direction (toolkit predicates in `rrs_geometry.py`).
- **Figures.** Overlay lines were computed in the viewer's own world coordinates and converted to screen space with the viewer's transform, then captured with the game frame underneath.

7. Limits and assumptions

- **Hull model of the rules engine.** The renderer's sprite is the best available evidence of hull geometry, but the engine's own overlap test may use different offsets. The finding is robust to a symmetric 25 u hull and reverses only for a hull about 8% longer than drawn.
- **Margin size.** 0.08 BL is about 0.26 s of travel at the boats' speed. Positions are exact simulator output at 1 Hz and the motion is smooth, so interpolation error is small, but this is a fine margin (about a foot on a Laser) and rule 18.2(e) is cited for that reason.
- **Contact.** The replay carries no hull polygons for collision; contact and rule 14 are not adjudicated.
- **Engine semantics.** The meaning of the INFRACTION timestamp (first breach versus contact) and of an EXONERATION without a penalty turn are undocumented.
- **Units.** The replay's unit is labelled metres but is not a real metre for the class modelled; the game's reported speeds (14 kt downwind) are likewise not Laser speeds. Distances are therefore reported in hull lengths. Confirm the intended unit scale with the author.
- **Clock offset.** The replay viewer's clock runs about 0.25 s ahead of the sample index; times in this report use the sample clock unless stated.

8. Revision history

> **Initial finding (superseded).** A first analysis assumed a 33.6 u hull, derived from the zone radius divided by three, and found Sully overlapped by 7.7 u at zone entry. A second draft of this report quoted game units as real metres; this version states distances in hull lengths. That assumption was flagged as decisive at the time. Measuring the hull directly from the game's renderer showed it to be 24 u with an asymmetric position point, which reverses the finding. The game's zone radius is not tied to hull length. The toolkit's default hull length should be corrected for this game.

9. References

1. World Sailing, *The Racing Rules of Sailing 2025–2028*. Part 2, Section C (rules 18–20): https://www.racingrulesofsailing.org/rules?part_id=141 . Definitions: https://www.racingrulesofsailing.org/definitions . Rules 11, 16.1 (Part 2, Sections A and B): https://www.racingrulesofsailing.org/rules?part_id=133 and ?part_id=136 . Rule 43 (Part 2, Section D).
2. World Sailing, *The Case Book 2025–2028*, v2025-07, Cases 2, 12, 25, 59, 63, 114, 118. https://media.sailing.org/sailing/wp-content/uploads/2025/07/31104846/WS-Case-Book-2025-2028-v2025-07.pdf
3. World Sailing, RRS Appendix M, *Recommendations for Protest Committees* (procedure: find facts, decide on the facts found, presume no breach until established, treat video and replay evidence with care). https://www.racingrulesofsailing.org/rules?part_id=159
4. Regattatron replay: https://regattatron.com/race/4ec6514d-62de-4858-a1b1-09b7c46e99e8/replay ; debrief: https://regattatron.com/race/4ec6514d-62de-4858-a1b1-09b7c46e99e8/debrief
5. ProtestCommittee toolkit: `PROTEST_COMMITTEE_METHODOLOGY.md`, `rrs_geometry.py`, `committee.py`; analysis scripts `overlap_check.py` and sample window `win_4ec6514d.jsonl` (session scratchpad).

10. Questions for the game's author

- What hull length, beam and bow/stern offsets does the rules engine use for overlap and zone tests? The renderer draws 24 × 10 units with the position point 14 units from the bow. What real length is one unit meant to represent?
- Does the INFRACTION timestamp mark first keep-clear breach or contact?
- What produced the simultaneous EXONERATION of AidanGoldfarb and INFRACTION of Sully at t = 2801.5 (automatic re-adjudication, a player protest, or something else)?
- Why is Sully's charge tagged rule 16 when she was never the right-of-way boat?